

Editorial

Systematic reviews: Guidelines, tools and checklists for authors

Are literature reviews obsolete? The immediate and obvious (albeit controversial to some) answer is yes, they are – they have been superseded by far better methods that add value and credibility to work that would otherwise lack objective measures of quality and transparency. Literature reviews have served the health professions for decades, but it is time to lay the traditional literature review to rest. Systematic reviews have long been established as the review type of choice. In this editorial we offer suggestions as to why literature reviews should be considered extinct, and outline recommendations for authors considering how to improve the quality of their review projects. This editorial presents a practical guide for *Nursing & Health Sciences* authors to increase the scientific rigor and validity of their submissions. The focus is on methods for qualitative and quantitative systematic reviews. Guidance exists for a wide range of synthesis methodologies, but qualitative and quantitative approaches are the two most commonly reported in *Nursing & Health Sciences*.

WHAT IS A SYSTEMATIC REVIEW AND HOW DOES THIS DIFFER FROM A TRADITIONAL LITERATURE REVIEW?

Systematic reviews aim to provide a comprehensive, unbiased synthesis of many relevant studies in a single document.
(Aromataris & Pearson, 2014)

Literature reviews are not guided by transparently reported methodology and methods, carry an increased risk of bias because of subjective decision-making, often undertake limited searching (or contain limited reporting), and tend to lack reproducibility (an important characteristic of good science) and transparency in decision-making – what was done, how it was done, and why those decisions were made (Aromataris & Pearson, 2014).

This editorial provides guidance and links to resources that will assist potential authors to align their methods to current expectations for methodology based on internationally recognized guidance for the planning and conduct of systematic reviews. These sources are the Joanna Briggs Institute (JBI, 2014) and Cochrane methodological guidance (Higgins & Green, 2011). Both are detailed, open access resources suited to the knowledge needs of the health professions that will assist authors to avoid common errors. Three of the most common errors in writing reviews for publication are the incomplete use of guidance, the incorrect use of reporting tools, and the incomplete use of reporting tools:

1. Use of incomplete guidance is common in literature reviews, particularly where authors describe a search as

based upon guidance from Cochrane but do not include a copy of their search strategy, fail to include adequate and appropriate key words and terms, or report incomplete or inadequate methods, and incorrectly cite the Cochrane handbook as the basis of their search.

2. The second critical error some authors make is to assume that including a checklist in their paper is the same as having conducted a systematic review. Checklists such as Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and Assessing the Methodological Quality of Systematic Reviews (AMSTAR) (see Table 1) indicate preferred reporting requirements, but are not detailed “how-to” guidelines (Shea *et al.*, 2007; Moher *et al.*, 2009). (PRISMA is a tool to structure the reporting of searching and study selection throughout the review [Moher *et al.*, 2009]) Incorrect use of reporting tools typically involves the use of the PRISMA flowchart for reporting of study selection choices and presenting this as evidence that the review was systematic. This is a misuse of the PRISMA flowchart, which was intended to illustrate decision-making related to study selection in systematic reviews and meta-analyses, but was never intended for method-free literature reviews. Similar issues have been noted with the use of critical appraisal instruments or standardized data extraction tools being included in traditional literature reviews, without following the full guidance for a systematic review.
3. Tools for improved reporting including Grading of Recommendations Assessment, Development and Evaluation (GRADE) and ConQual (see Table 1) are useful for confirming or reporting specific parts of a review, but single tools are not to be confused with complete methodological guidance suited to publication in an international peer reviewed journal (Guyatt *et al.*, 2008; Munn *et al.*, 2014). GRADE, for example, is an approach to transparent reporting of the strength of findings, and relates to communicating the findings, rather than detailed methodological guidance for the full systematic review (Guyatt *et al.*, 2008).

METHODOLOGICAL GUIDANCE

An overview of the methodological guidance available for different types of review questions is listed in Table 2. Guidance from groups such as the Centre for Reviews and Dissemination (CRD), Cochrane, EpPI-Centre, the Agency for Healthcare Research and Quality (AHRQ), reporting initiatives in the EQUATOR network, and the Institute of Medicine (IOM) is also readily available. Methodological guidance extends well

Table 1. Common checklists and tools for assessing or reporting steps or processes within a systematic review

Purpose	Item	Quantitative	Qualitative
Checklist for reviews of effects with or without meta analysis. Note there is a companion document that provides guidance for each item in the checklist.	PRISMA	Reviews of effects and meta-analysis	No
Generic assessment instrument	ENTREQ	No	Selected qualitative methodologies
To assess evidence or develop recommendations	GRADE	Yes	No
Rating the confidence of qualitative systematic review findings	ConQual	No	Selected qualitative methodologies
Assesses reporting	COREQ	No	Selected qualitative methodologies

COREQ, Consolidated Criteria for Reporting Qualitative Research; ENTREQ, Enhancing transparency in reporting the synthesis of qualitative research; GRADE, Grading of Recommendations Assessment, Development and Evaluation; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

Table 2. Methodological guidance for conducting systematic reviews

Purpose	Item	Quantitative	Qualitative
Detailed methodological guidance for systematic reviews: Effectiveness Meaningfulness Costs Prevalence/incidence Diagnostic accuracy Etiology and risk Policy Mixed methods Umbrella reviews Scoping reviews	JBIR Reviewer's Manual and JBISRR Author guidelines. Recommended for use with SUMARI 1.0 software for synthesis of diverse types of evidence. http://joannabriggs.org/sumari.html	Yes, reviews that seek to generate recommendations for policy or practice	Qualitative reviews that seek to generate findings that describe, explain or inform decision making for policy and practice
Detailed methodological guidance for systematic reviews of effectiveness (including integration of qualitative data), diagnostic test accuracy, individual patient data reviews, overviews and reviews in public health and health promotion	Cochrane Handbook and MECIR . Recommended for use with Review Manager 5.3 or above for quantitative evidence. http://handbook.cochrane.org/	Yes, reviews that seek to generate findings for policy or practice	Qualitative reviews that seek to generate higher order theories for testing

JBISRR, Joanna Briggs Institute Database of Systematic Reviews and Implementation Reports; MECIR, Methodological Expectations of Cochrane Intervention Reviews.

beyond the scope and utility of a checklist or reporting guide, it provides detailed, practical guidance for each step, process, and decision required when writing a systematic review (Higgins & Green, 2011; Joanna Briggs Institute, 2014).

PLANNING THE SYSTEMATIC REVIEW

Both JBI and Cochrane require that systematic reviews (as with primary research) are guided by an a priori protocol (Higgins & Green, 2011; Joanna Briggs Institute, 2014). This means planning the topic and methods to suit the question before commencing the project. Having the methods written up and published in a protocol means many of the decisions that need to be made are recorded before the review results are known, which reduces the risk of bias when making decisions regarding which papers to include, which outcomes to measure, and which analytic techniques will be used to

report the review results. We would not trust the results of research where the researchers did not provide a detailed report of their methods, where they highlighted results that were not accurate, or published conclusions that went beyond that indicated by the data. Nor should we accept reviews that lack evidence of a clear methods section and an audit trail that demonstrated the protocol was adhered to.

If the protocol is written according to high quality guidance, systematic review authors are less likely to miss key steps, make incorrect decisions, or use tools and checklists incorrectly. In Cochrane and JBI guidelines, the information required when writing a protocol also make up the steps to complete a systematic review, and include a description of:

1. A structured question and objective
2. Detailed inclusion and exclusion criteria, (these are often based upon the question structure)

3. A comprehensive search to identify all relevant studies (published and unpublished)
4. Critical appraisal of the included studies
5. Data extraction
6. Analysis of data extracted from the included research studies
7. Presentation of the findings

Step 1: Developing a structured question

Detailed guidance on where and how to search for different types of evidence in the JBI Handbook, including advice for specific databases and how to integrate PRISMA reporting should be combined with expert advice from an information scientist, and training for reviewers.

The PRISMA checklist is helpful for identifying the minimum standards for a protocol. The PRISMA flow chart reports study selection and inclusion decisions.

To assist in structuring a well designed question, use of the PICO mnemonic, or the qualitative variant PICo, is encouraged. In quantitative reviews the PICO mnemonic includes a description of the population, the intervention, a comparator and primary outcome of interest; and in qualitative reviews the PICo mnemonic includes the population, phenomena of interest, and context. In a 2014 systematic review, Kesänen *et al.* (2014) questioned the evidence on use of knowledge tests in patient education.

Step 2: Inclusion criteria

The inclusion criteria expands on the question, enabling readers to understand which studies were likely to be considered for inclusion and why. For example, Kesänen *et al.* (2014) specified that studies would only be included if the **population** was patients, where the education was for a specific health problem, where the **intervention** was health education that included a knowledge test, with a **comparator** intervention (another patient education program or standard care), that the **study designs** be either randomized or quasi-randomized, and the **outcomes** reported data on measures of patient knowledge specific to their illness.

Step 3: The search strategy

A detailed search strategy is described, along with the key terms that guided the search (Kesänen *et al.*, 2014). The review reports language (English) and date limits (Jan 2000–Feb 2012), as well as a process for hand searching of reference lists – indicating that the authors were familiar with guidelines for reporting searching rather than simply relying on the PRISMA flow chart to illustrate the numbers of studies throughout the review. Being able to read the search strategy, the key words, and exclusion criteria and comparing these with the title enables a reader to independently decide whether the search was both comprehensive and appropriate to the topic. If

the reporting of a search strategy fails this basic test, the entire paper and its findings may be questioned for completeness, accuracy, and reliability. When searching, follow Cochrane and JBI guidance to ensure the following are met as minimum criteria for a comprehensive and exhaustive search:

- Identify all relevant synonyms for each key word and each database
- Integrate key words and controlled vocabulary
- Use Boolean operators to manage search strings and combine concepts
- Provide a detailed report including the date the search was conducted per database as well as any date or language limits in the search.
- Utilize the skills of librarians and information scientists in ensuring an appropriate search strategy that is a balance of sensitivity (large numbers of results) and specificity (most relevant results).
- JBI also recommends that a copy of at least one entire database search be appended to the review

The guidelines for these search strategies are used with the PRISMA flow chart. What makes these sources of international guidance valuable is the descriptions of “how,” “when,” and “why” to implement a high-quality search that can be reported using the PRISMA flow chart.

Step 4: Critical appraisal

Establishing the rigor or trustworthiness of research studies must include reporting why any studies were excluded.

The process of identifying which studies should be considered for inclusion is based upon detailed inclusion criteria related to the population, the intervention (or phenomena) of interest, and the study type. Once the relevant studies have been identified, they should be evaluated for quality. Often referred to as critical appraisal, appraisal should focus on internal validity (for quantitative papers) or trustworthiness (for qualitative papers). Use of a known tool or instrument to guide reviews in applying a standardized approach is common: both Cochrane and JBI have integrated standardized study quality assessment in their systematic review software (Higgins & Green, 2011; Joanna Briggs Institute, 2014). Additional support for methodological quality assessment is readily available (Pearson *et al.*, 2014). The guidance directs users in how to assess studies, including piloting, establishing consensus, and how to report appraisal decisions. Reviewers using these approaches give readers clear knowledge of the quality of the included studies by reporting quality criteria most relevant to the study type and review objectives.

Step 5: Data extraction

For data extraction, the authors describe how they used checklists. Processes exist for a wide range of study designs. The explanation is a critical piece of information that assists the reader to evaluate what decisions were made during data

extraction, as well as how and why they were made. The specific data extracted can then be compared with the inclusion criteria to validate the reliability of the project. Data extraction commonly includes the details of the citation (author/s, title, year, journal of publication), but also includes two other types of data. General information is extracted on the study design, the geographic location, and important characteristics associated with how the study was conducted. Cochrane and JBI guidance establishes minimum reporting criteria as including:

- The study methods
- Participant characteristics or demographics
- The interventions, including number of groups and participants per group, and comparisons (such as active or passive placebo, or usual care). (Qualitative reviews) the Phenomena of Interest – the experiences people have in relation to an interest
- The outcome definition, how the outcome was measured and how the unit/s of measurement should be interpreted. (Qualitative reviews) the Context
- Results including the numbers of participants allocated, the number treated, and the sample size. (Qualitative reviews) themes, metaphors, analytic data, and related participant quotes

The second type of data collected is “outcome data.” In a review of the effectiveness of an intervention, outcome data will be statistical: measures such as relative risks, odds ratios, or weighted means and standard deviations are extracted for each outcome along with the confidence interval. If the steps above are taken by reviewers who are familiar with and follow guidance, the paper submitted for publication will be of significantly better quality and reliability than a traditional literature review.

In qualitative reviews, the nature of the data varies, as text and participant quotes become the primary data of interest (Lockwood *et al.*, 2015). It is important that when extracting data from qualitative papers the extract is verbatim, not edited or abridged. Data for qualitative synthesis is described as “analytic data” and may include primary study author conclusions, opinions, or results of their own analysis, such as themes (Joanna Briggs Institute, 2014). Each piece of data extracted is called a finding. Where possible, each finding should also have a quote from a study participant to verify the linkage between the authors published results and the participant’s words. In the JBI guidance, a measure of credibility is allocated to each extracted “finding,” and this facilitates the final assessment of rigor in the final assessment of confidence in the qualitative synthesis results. These findings are then synthesized either based on similarity in meaning leading to recommendations for policy and practice (JBI), or through an inductive approach that leads to the creation of new theories for testing (Cochrane) (Lockwood & Pearson, 2013).

Step 6: Analysis of data extracted from the included research studies

The Cochrane Handbook outlines reporting criteria for meta-analysis and how to avoid common errors.

Knowledge of correct methods and decision-making in the analysis of data extracted from the included research studies requires training and statistical advice or support. In this section, we focus instead on minimum reporting requirements for meta analysis and meta-synthesis. Meta-analysis is the statistical combination of data from two or more studies resulting in a pooled effect measure (Higgins & Green, 2011). In any review that includes meta-analysis, additional data must also be reported so that readers are able to see the effect measure, its significance, and its strength. Authors who do not include confidence intervals when reporting *P* values, or misinterpret the measures of heterogeneity are making commonly identified errors, but errors are easily avoided by following guidance. The two most common errors authors make are: (i) to use a random effects model and present the resulting meta-analysis as statistically and clinically significant while ignoring or under-reporting heterogeneity, and (ii) to misinterpret the risk of heterogeneity represented by I^2 (I^2 is the preferred measure for heterogeneity).

Both JBI and Cochrane reporting requirements for meta-analysis include a description of the analytic model, effect measure, the use of double data entry, reporting of effect sizes, confidence intervals with *P* values, and measures of heterogeneity for each pooled outcome. When a sufficient number of studies is included, analysis of the risk of publication bias is also considered a requirement for good reporting. Papers that include meta-analysis should report all the above data for each outcome, otherwise the reporting is incomplete. A *P* value without confidence intervals is not helpful data, similarly, a pooled estimate of effect is incomplete if the measure of heterogeneity is not provided (Higgins & Green, 2011).

GRADE measures within-study risk of bias (methodological quality), directness of evidence, heterogeneity, precision of effect estimates, and risk of publication bias.

Step 7: Presentation of the findings

Cochrane and JBI follow slightly different formats at this point, but the basis of their guidance is to optimize clarity of reporting and accessibility of the findings. Both groups endorse the use of GRADE for quantitative recommendations (Guyatt *et al.*, 2008). GRADE provides a summary of the findings balanced against the extent to which an estimate of effect or association is close to the quantity of specific interest (Guyatt *et al.*, 2008). GRADE measures within-study risk of bias (methodological quality), directness of evidence, heterogeneity, precision of effect estimates, and risk of publication bias. In qualitative systematic reviews, JBI advocate for the presentation of the synthesis in a graphical form (i.e. a figure), and have software that facilitates both the conduct and reporting of the synthesis (Joanna Briggs Institute, 2014). A summary of findings table can be either quantitative or qualitative. Summary of findings tables improve the

presentation of outcome data, and related information that helps a reader make sense of the findings.

Quantitative summary of findings tables combine the outcome measure, a comparative risk measure, and the effect size (either as an odds ratio, relative risk, or mean difference), along with the number of study participants and the GRADE quality score in table format. A qualitative summary of findings table incorporates and reports the title, population, phenomena of interest, context, and for each final synthesized finding, the type of data it was based upon (either qualitative research or text), a dependability score, a credibility score, and a confidence rating that reflects the type of data, as well as the dependability and credibility scores (Munn *et al.*, 2014).

CONCLUSIONS

This editorial has focused on guidance based upon international guidance, tools, and checklists for the optimal conduct and reporting of systematic reviews. *Nursing & Health Sciences*, as an international peer reviewed journal, is committed to the highest standards for publication of all types of research, scholarly papers, and systematic reviews. The author guidelines available at: [http://onlinelibrary.wiley.com/journal/10.1111/\(ISSN\)1442-2018/homepage/ForAuthors.html](http://onlinelibrary.wiley.com/journal/10.1111/(ISSN)1442-2018/homepage/ForAuthors.html) should be closely read and followed before commencing writing for publication. *Nursing & Health Sciences* endorses the Committee on Publication Ethics (COPE), the standard by which papers submitted to *Nursing & Health Sciences* will be considered. Familiarity with the author guidelines is therefore a critical step in planning for publication.

If the protocol is based on high quality guidance, the resulting systematic review will be of higher quality than a literature review, it will likely have greater validity, create opportunities to network with groups such as Cochrane or JBI, and open doors to conference presentations or posters, as well as being congruent with the expectations of *Nursing & Health Sciences*, highlighting your work among the many submissions received each month. Systematic reviews are ideal projects for postgraduate students in Masters, PhD, or Doctoral programs, and lead to important publication opportunities. When used correctly, checklists and flowcharts

strengthen the quality and clarity of reporting, and can only add to the value of the knowledge being presented.

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